

Applied Linear Algebra (EE 635): Problem Set 2

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Questions

1. Describe all possible 3×3 elementary row operation matrices. Show that each of these matrices is invertible by explicitly constructing the inverse.
2. Find the kernel of the following matrices.
 - (a) $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$
 - (b) $A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{pmatrix}$
 - (c) $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix}$
 - (d) $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$
 - (e) $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$
3. For the matrices given in Question 2, find a matrix B such that $\ker(A) = \text{im}(B)$.
4. Let $A \in \mathbb{R}^{m \times n}$ with $m < n$. Show that there exists a **non-zero** vector $x \in \mathbb{R}^n$ such that $Ax = 0$.
5. A steam plant burns two types of coal: anthracite (A) and bituminous (B). For each ton of A burned, the plant produces 27.6 million Btu of heat, 3100 grams (g) of sulfur dioxide, and 250 g of particulate matter (solid-particle pollutants). For each ton of B burned, the plant produces 30.2 million Btu, 6400 g of sulfur dioxide, and 360 g of particulate matter.

- (a) How much heat does the steam plant produce when it burns x_1 tons of A and x_2 tons of B?
- (b) Suppose the output of the steam plant is described by a vector that lists the amounts of heat, sulfur dioxide, and particulate matter. Express this output as a linear combination of two vectors, assuming that the plant burns x_1 tons of A and x_2 tons of B.
- (c) Over a certain time period, the steam plant produced 162 million Btu of heat, 23,610 g of sulfur dioxide, and 1623 g of particulate matter. Determine how many tons of each type of coal the steam plant must have burned. Include a vector equation as part of your solution.
6. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be points in $\mathbb{R}^3$ and suppose that for $j = 1, \dots, k$, an object with mass m_j is located at point $\mathbf{v}_j$. The total mass of the system of point masses is:

$$m = m_1 + \dots + m_k$$

The center of gravity (or center of mass) of the system is:

$$\bar{\mathbf{v}} = \frac{1}{m} [m_1 \mathbf{v}_1 + \dots + m_k \mathbf{v}_k]$$

Compute the center of gravity of the system consisting of the following point masses (See Figure 1):

Point	Coordinates	Mass
$\mathbf{v}_1$	(5, -4, 3)	2 g
$\mathbf{v}_2$	(4, 3, -2)	5 g
$\mathbf{v}_3$	(-4, -3, -1)	2 g
$\mathbf{v}_4$	(-9, 8, 6)	1 g

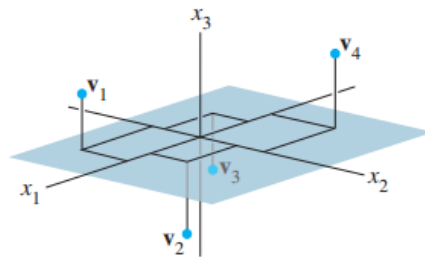


Figure 1